NATIONAL UNIVERSITY OF LESOTHO

Department of Mathematics and Computer Science

CS4433 Software Engineering: Final Mini Project

REQUIREMENTS ENGINEERING

STUDENT NUMBER: 202321162

STUDENT NAME: LEHLOHONOLO

STUDENT SURNAME: MAJORO

COURSE: CS 4433

PROJECT: LEARNING MANAGEMENT SYSTEM

PHASE 2: Stakeholder Analysis, Software Requirements Specification and Use Case Model

1. Introduction

1.1 Purpose and scope of this document

The school wants a Learning Management System in which lecturers set and mark assignments, a Head of Department (HoD) moderates the marks, and students see a mark only after it has been released. Phase 1 (project management plan) established that the project is feasible within about 214 hours over seven weeks. This Phase 2 document defines **what** the system must do, so that design (Phase 3) and implementation (Phase 4) can be checked against a stable, agreed baseline at Gate 1.

1.2 Method used

The five stakeholder statements and the case study were analysed in four steps:

- Decompose. Each statement was split into single, checkable needs (N1–N15, Section 2.3).
- Compare. Needs were compared against each other and against the case study to find direct conflicts, vague wording and missing information.
- Resolve. Each problem was resolved by applying, in order: legal and security obligations, the case study text, the policy owner's authority (Principal, then HoD), fairness to students, and feasibility within the 214-hour budget. Every resolution is written as a numbered business rule (BR-xx).
- Specify. Business rules were converted into verifiable requirements (FR-xx, NFR-xx) and then grouped into use cases (UC-xx), so every requirement can be traced back to a stakeholder need.

2. Stakeholder Analysis

2.1 Stakeholder identification

- Five stakeholders were interviewed. Four more were identified from the case study and from the legal context. The administrator and the regulator matter because they were not interviewed, so their needs have to be inferred.

ID	Stakeholder	Type	Main interests	Main concerns	Influence / Interest
S1	Principal	Sponsor, policy owner	A simple system teachers can use; no mark visible to a student before the HoD has checked it.	Unchecked marks reaching students; system too complex for non-IT staff.	High / High
S2	Lecturers	Primary user	Mark quickly; give feedback; give extensions to students with good reasons; students receive marks soon after marking.	Waiting for moderation; extra paperwork; losing control of extensions.	Medium / High
S3	Heads of Department	Primary user, moderator	Consistent late-penalty rule; see every mark in the department before release; release approved marks.	Large moderation workload; inconsistent marking; disputes about penalties.	High / High
S4	Students (represented by the student representative)	Primary user	Fix mistakes and resubmit before the deadline; written feedback; late work not rejected; fair penalties.	Work being refused; marks seen late; unfair or unclear penalties.	Low / High
S5	IT officer	Technical authority	Offline operation on old computers; protected passwords; fast response with 800 students.	Slow software; stolen or leaked passwords; dependence on the internet.	Medium / High
S6	System administrator	Primary user (not interviewed)	Easy management of accounts, departments, courses, enrolment.	Errors in enrolment; accidental deletion of data.	Medium / Medium
S7	Parents and guardians	Indirect	Their children's personal data is kept private; marks are fair.	Disclosure of minors' information.	Low / Medium

ID	Stakeholder	Type	Main interests	Main concerns	Influence / Interest
S8	Data protection authority (Data Protection Act, 2011)	External, legal	Lawful, minimal, secure processing of personal data, including minors' data.	Excess data collection; weak security; no accountability for changes.	High / Low
S9	Project supervisor	Oversight	Complete, consistent, justified artefacts at each gate.	Gaps and inconsistency between artefacts.	High / High

3. Conflicts and Ambiguities

Seven conflicts (two or more stakeholders want incompatible things) and fourteen ambiguities or omissions (a statement can be read in more than one way, or leaves something undefined) were found. Each is resolved below. The resolutions are the agreed interpretation of the requirements; if a stakeholder disagrees at Gate 1, the change-control procedure in the project plan applies.

3.1 Conflicts

ID	Needs in conflict	Analysis	Resolution and justification
CF1	Principal, HoD vs Lecturer: when do students see marks?	The lecturer wants students to see marks immediately after marking; the Principal and HoD forbid any unchecked mark from reaching students. Both cannot be fully satisfied.	Marks are released in two stages. The lecturer sees and edits the mark as soon as it is saved; the student sees it only after the HoD has moderated and released it (BR-02). The case study itself gives the HoD the power to release marks, and the Principal owns policy, so N2 prevails. To protect N3: the HoD can approve and release a whole assignment in one action, and lecturers can see the count of marks awaiting moderation, so delay is visible. No automatic release by timeout is allowed because that would defeat N2.

ID	Needs in conflict	Analysis	Resolution and justification
CF2	Lecturer extension vs HoD late penalty	If a lecturer grants more time, the work could still be penalised as “late” against the original deadline, which would make extensions meaningless.	Effective deadline. Each student has an effective deadline = the later of the assignment deadline and any extension granted to that student. Lateness and penalties are always calculated against the effective deadline, so work inside an extension is not late. Extensions need a recorded reason so the HoD can see how the lecturer used the power.
CF3	late work must not be rejected vs zero after three days	“Zero after three days” could be read as refusing the work. The student representative says late work must not be rejected.	Accept, flag, and mark. The system always accepts late submissions. After more than three days the submission is stored and flagged, and its final mark is zero. The lecturer can still mark and give feedback, and only the HoD can override the zero, with a recorded reason. This satisfies both statements literally, and the student keeps a dated receipt as evidence.
CF4	Principal: keep it simple vs the full set of role rules, audit log and moderation	The case study demands four roles, a submission life cycle, moderation and an audit log. Teachers are not IT experts.	Complexity is hidden from the user. Menus show only the actions of the user’s role, use at most three menu levels, plain-language prompts, and confirmation before saving a mark. The penalty, audit record and status change are automatic. Nice-to-have features are classed Should or Could and are cut first.
CF5	HoD sees every mark vs workload for 800 students	Reviewing every mark individually may not be practical, but the HoD requires full visibility and the Principal requires a check.	The HoD can list all marks for a course and assignment with raw mark, penalty, final mark and feedback and may approve individually or in batch. Batch approval still requires an explicit action and a count confirmation, so no mark is released unseen.

ID	Needs in conflict	Analysis	Resolution and justification
CF6	resubmit freely vs fair marking and deadline	Unlimited resubmission could change work after a lecturer has begun marking, and resubmitting after the deadline would bypass the penalty.	A student may replace a submission only while the effective deadline has not passed and the work has not been marked. The latest version is the one marked; earlier versions are kept for the record. A student who has submitted late cannot resubmit, since the deadline has already passed.
CF7	strong password protection vs old computers, fast response	A deliberately slow, memory-hard password hash gives strong protection but may slow login on old hardware.	Tune, do not weaken. A recognised salted hash will be used with the library's interactive cost parameters, tuned so login stays within 2 seconds on the reference machine with 800 students stored. Only one hash is computed per login, so the number of stored users does not affect login time.

4. Software Requirements Specification

4.1 Introduction

4.1.1 Purpose

This SRS states the requirements of the Moderated LMS so that the designer, developer, tester and school stakeholders share one description of the system.

4.1.2 Scope

The product is a C++ console application that lets a secondary school manage courses, assignments, submissions, marking, moderation and release of marks for four roles, with persistent storage in My SQL and an audit log of every mark change. It runs offline on a single school computer or a local network.

4.1.3 Definitions

Term	**Meaning**
Raw mark	The mark entered by the lecturer before any late penalty.

Term	Meaning
Final mark	Raw mark minus late penalty (not below 0), possibly adjusted by the HoD. This is the mark the student sees after release.
Effective deadline	The deadline that applies to one student for one assignment.
Moderation	The HoD's check of a lecturer's mark: approve, adjust or return.
Release	The HoD action that makes a Moderated mark visible to the student.
Submission	One student's work for one assignment; it may have several versions before the deadline.
Audit record	An unalterable log entry of a mark change: who, when, old value, new value, reason.

4.2 Overall Description

4.2.1 Product perspective

The LMS is a stand-alone application. It has no connection to other school systems and no internet dependency. It will be built in layers (presentation, business logic, data access) over a local MySQL database file.

4.2.2 Product functions

Role	Main functions
All users	Log in, change password, be logged out after inactivity.
Administrator	Manage accounts, departments, courses, lecturer assignment and student enrolment; reset passwords; back up data.
Lecturer	Create and edit assignments, grant individual extensions, view submissions, mark with feedback, re-mark before moderation.
Head of Department	Review all marks of the department, moderate, release, correct released marks, view the audit log.
Student	View courses and assignments, submit, resubmit, submit late with penalty, view released marks and feedback.

4.2.3 User characteristics

User	Characteristics that affect the design
Lecturers, HoDs	Competent with office software but not IT experts (N1): menu-driven, plain-language prompts, no commands to memorise.
Students	Secondary school learners (minors): simple screens, clear confirmation of what was submitted, and privacy of marks.

User	Characteristics that affect the design
Administrator, IT officer	More technical; install, back up and restore the system using the installation guide.

4.2.4 Constraints

- C++, My SQL database accessed only through a data-access layer, parameterised queries only .
- A recognised library for salted password hashing.
- Console interface (or a simple desktop GUI is allowed by the brief, but the console is chosen for the old computers and the 7-week schedule).
- Old computers, unreliable internet: no network dependency.
- Lesotho Data Protection Act, 2011: collect only necessary data and protect it.
- One developer, about 210 available hours.

4.2.5 Assumptions and dependencies

- The school provides the initial lists of users, departments, courses and enrolments to the administrator.
- The computer's clock is correct, because deadlines and penalties depend on it.
- Each department has one HoD who does not mark work in the same system under the same account.

4.3 Specific Requirements

4.3.1 External interface requirements

Interface	Requirement
User interface	Text menus and prompts on a console; at most 3 menu levels; each prompt states what to enter and the allowed range; destructive or final actions ask for confirmation.
Software interface	SQLite database file, accessed only by the data-access layer with parameterised statements; libsodium for password hashing; no other network or external software.
File system	Submitted files are copied to a managed folder; the application reads a file path entered by the student and writes only to the managed folder and to export or backup folders chosen by authorised users.
Hardware	Standard keyboard, monitor and hard disk of the reference machine. No special devices.
Communications	None. The system must work with all network interfaces disabled .

4.3.2 Functional requirements

Every requirement has a unique identifier, a MoSCoW priority, its source (stakeholder need N, conflict CF, ambiguity AM, business rule BR, or the case study) and a verification method. Each is written with numbers, ranges or observable behaviour so that it can be tested.

Table 4.1: Functional requirements

ID	Requirement	Priority
FR-01	The system shall require a username and password before any function is available, and shall then show only the menu for the user's role (Administrator, Lecturer, HoD or Student).	Must
FR-02	On a failed login the system shall show the same message ("Invalid username or password") whether the username or the password was wrong. After 5 consecutive failed attempts for one account it shall refuse logins for that account for 5 minutes.	Must
FR-03	The system shall allow each role to execute only the functions shown for it in the permission matrix (Table 4.5). An attempt to use any other function shall be refused with a message and recorded in an event log.	Must
FR-04	A user shall be able to change their own password by entering the current password and a new one of at least 8 characters containing a letter and a digit. A temporary password set by the administrator shall have to be changed at the next login.	Must
FR-05	The system shall log a user out automatically after 10 minutes without input and return to the login screen without saving any half-finished action.	Should
FR-06	The administrator shall be able to create, edit and deactivate user accounts. Each account has a unique username, full name, role and, as applicable, a department (lecturer, HoD) or student number (student). Accounts shall not be deleted, only deactivated; a deactivated user cannot log in.	Must
FR-07	The administrator shall be able to reset a user's password to a temporary one.	Should
FR-08	The administrator shall be able to create and edit departments and courses. Each course has a unique code and belongs to exactly one	Must

ID	Requirement	Priority
	department; each department has at most one HoD.	
FR-09	The administrator shall be able to assign one or more lecturers to a course and remove them.	Must
FR-10	The administrator shall be able to enrol and unenrol students in a course. Unenrolling shall not delete existing submissions or marks.	Must
FR-11	A lecturer shall be able to create an assignment for a course they teach, giving a title, description, deadline (date and time, in the future) and maximum mark (integer from 1 to 100).	Must
FR-12	A lecturer shall be able to edit the title and description of an assignment, move its deadline later (never earlier), and close it so that it accepts no more submissions. The maximum mark shall not be editable once a submission exists.	Should
FR-13	A lecturer shall be able to grant one student an extension on one assignment by entering a new deadline later than the student's current effective deadline and a reason of at least 10 characters. The system shall record the lecturer, the time and the reason, and shall use the new deadline as that student's effective deadline.	Must
FR-14	A student shall be able to list their enrolled courses and the assignments in each, showing for each assignment the deadline, the student's effective deadline if different, and the status of the student's own submission.	Must
FR-15	A student shall be able to submit one file for an assignment in an enrolled course. The system shall store a copy of the file, record the time, file name, size and version number, set the submission status (Submitted if on time, Late if after the effective deadline) and display a receipt.	Must
FR-16	The system shall reject a file that does not exist, is empty, is larger than 10 MB, or is not of type PDF, DOCX, TXT, ZIP, PNG or JPG, shall state the reason, and shall leave any existing submission unchanged.	Must

ID	Requirement	Priority
FR-17	A student shall be able to replace their submission any number of times while the effective deadline has not passed and the submission has not been marked. The latest version shall be the one marked; earlier versions shall be kept.	Must
FR-18	The system shall accept a submission made after the effective deadline, mark it Late, warn the student of the days late and the penalty and ask for confirmation before saving, and shall not refuse it for lateness. A student who has submitted late shall not be able to resubmit.	Must
FR-19	When a lecturer marks a submission, the system shall automatically calculate days late and the late penalty by BR-05 and BR-06 (10% of the maximum mark per day for 1–3 days; final mark zero for 4 or more days), and shall display raw mark, penalty and final mark.	Must
FR-20	A lecturer shall be able to list the submissions for the assignments of their courses, with student, time, status and days late, and to open the stored file of a submission.	Must
FR-21	A lecturer shall be able to mark a submission by entering a raw mark between 0 and the maximum mark (steps of 0.5) and written feedback of 10–2000 characters. Saving sets the status to Marked.	Must
FR-22	A lecturer shall be able to change the mark or feedback of a submission while its status is Marked or Returned, entering a reason of at least 10 characters. A lecturer shall not be able to change a submission that is Moderated or Released.	Must
FR-23	A HoD shall be able to list all submissions for the courses of their own department, filtered by course, assignment and status, showing raw mark, days late, penalty, final mark, lecturer and feedback.	Must
FR-24	A HoD shall be able to moderate each Marked submission by choosing Approve (status Moderated), Adjust (enter a new final mark between 0 and the maximum, with a reason of at least 10 characters; status Moderated) or Return	Must

ID	Requirement	Priority
	for remarking (enter a comment; status Returned). A HoD shall not be able to moderate a mark that they entered themselves.	
FR-25	A HoD shall be able to approve all Marked submissions of one assignment in one action after viewing the number of submissions affected and confirming.	Should
FR-26	A HoD shall be able to release Moderated submissions, individually or for a whole assignment. Release sets the status to Released and records the release time. No other action shall release a mark.	Must
FR-27	A student shall see the mark, penalty and feedback of a submission only when its status is Released. For other statuses the student shall see only the status (Submitted, Late, or “Awaiting release”) and no mark or feedback.	Must
FR-28	The system shall allow only the status changes in the life-cycle table (Table 4.6) and shall refuse any other change with a message.	Must
FR-29	Every creation or change of a mark (lecturer marking, lecturer re-marking, HoD adjustment or penalty override, post-release correction) shall store an audit record with who, when, old value, new value and reason, in the same database transaction as the change. The application shall provide no function to edit or delete audit records.	Must
FR-30	A HoD shall be able to view the audit records for the courses of their department, filtered by assignment, student or date range.	Must
FR-31	The system shall validate all input (type, range, length, allowed values). On invalid input it shall show a message and ask again. On a database or file error it shall roll back the operation, show a message and return to the menu without terminating.	Must
FR-32	On login, a student shall be told how many marks were released since their last login.	Could

ID	Requirement	Priority
FR-33	A lecturer or HoD shall be able to export the released marks of an assignment to a CSV file.	Could
FR-34	The administrator shall be able to create a backup copy of the database and the stored submissions in a folder chosen by the administrator.	Should
FR-35	A HoD shall be able to correct the mark of a Released submission by entering a new final mark and a reason of at least 10 characters. The status shall remain Released and an audit record shall be written.	Should

4.3.3 Non-functional requirements

Each non-functional requirement states a measurable limit and the way it will be measured.

Table 4.2: Non-functional requirements

ID	Category	Requirement (measurable)	Priority
NFR-01	Performance (login)	Login shall complete within 2 seconds, measured from pressing Enter on the password to the display of the menu, with 800 student and 50 staff accounts and 20,000 submission records stored, on the reference machine (dual-core 2 GHz CPU, 4 GB RAM, hard disk). All of 30 consecutive timed runs shall meet the limit.	Must
NFR-02	Performance (operations)	Every menu operation shall display its result within 1 second, except listing up to 500 submissions for the HoD, which shall take no more than 2 seconds, with the same data volume and machine as NFR-01 (measured as the 95th percentile of 50 runs).	Must
NFR-03	Resource use	The application shall run on the reference machine with a peak memory use of no more than 150 MB and an installed size (excluding submitted files) of no more than 50 MB.	Should
NFR-04	Offline operation	All functions shall work with every network interface disabled. The application shall make zero outgoing network connections (checked by monitoring connections while running the full system test suite).	Must
NFR-05	Security (passwords)	100% of stored passwords shall be salted hashes made with Argon2id (libsodium), with a unique	Must

ID	Category	Requirement (measurable)	Priority
		random salt of at least 16 bytes per user. No password, in any form other than the hash, shall appear in the database, in log files or in error messages (checked by inspecting the database and logs after the test suite).	
NFR-06	Security (injection)	100% of database statements shall use parameterised queries; no user input shall be joined into SQL text. At least 10 injection strings (e.g. ' OR '1'='1' --) entered in every text field shall have no effect on the data.	Must
NFR-07	Security (access control)	In the role-by-function test, 0 out-of-role functions shall be reachable or executable, and 100% of permitted functions shall work.	Must
NFR-08	Robustness	The application shall not crash or terminate abnormally in any of at least 100 prepared invalid-input cases and at least 5 simulated failures (read-only database, locked database, missing submission file, full disk, unreadable file).	Must
NFR-09	Data integrity	A mark change and its audit record shall be committed together or not at all. Over at least 20 fault-injection runs there shall be 0 cases of a changed mark without an audit record. Foreign-key checks shall be enabled in every database connection.	Must
NFR-10	Usability	No function shall be more than 3 menu levels from the main menu. Using only the user manual, each of at least 3 volunteers acting as lecturers shall create an assignment and mark a submission within 5 minutes each.	Must
NFR-11	Accessibility	All information shall be given as plain text (no meaning conveyed only by colour or position), so that a screen reader can read it. One complete user scenario per role shall be completed using Windows Narrator or NVDA.	Should
NFR-12	Maintainability	The business-logic layer shall have at least 70% line coverage from automated tests. The static analyser (cppcheck or clang-tidy) shall report 0 errors and 0 unaddressed warnings. At least 90% of functions shall have cyclomatic complexity of 10 or less and none above 15.	Must

ID	Category	Requirement (measurable)	Priority
NFR-13	Portability	From a clean checkout, the project shall build with a documented CMake command sequence and a C++ compiler on Windows 10/11 and on Ubuntu 22.04, and all tests shall pass on both.	Should
NFR-14	Privacy (data minimisation)	The database shall hold no personal data other than full name, username, student number, role, department and password hash (checked against the data dictionary and the schema).	Must
NFR-15	Recoverability	A school technician following the installation guide shall be able to restore the system from a backup made with FR-34 in no more than 10 minutes, with all marks and audit records intact.	Should

4.3.4 Business rules

The business rules are the agreed resolutions of Section 3 in compact form. Requirements refer to them so that a rule is stated once.

Table 4.3: Business rules

ID	Rule
BR-01	Each account has exactly one role. A person who needs two roles has two accounts.
BR-02	A student may see a mark and its feedback only when the submission status is Released.
BR-03	Effective deadline = the later of the assignment deadline and the student's extension deadline (if any).
BR-04	A submission is late if its time is later than the effective deadline. A submission exactly at the deadline is on time.
BR-05	Days late d = ⌈(submission time − effective deadline) ÷ 24 hours⌉ in calendar days, for late submissions; d = 0 otherwise.
BR-06	Penalty = 10% × maximum mark × d for d = 1, 2 or 3. If d ≥ 4 the final mark is 0. Final mark = max(0, raw mark − penalty). See Table 4.4.
BR-07	The lecturer cannot waive a penalty. A submission with d ≥ 4 is still marked and given feedback by the lecturer. Only the HoD can adjust the final mark, with a reason.
BR-08	Resubmission is allowed only while now ≤ effective deadline and the status is Submitted. The latest version is marked; earlier versions are kept.

ID	Rule
BR-09	Only a lecturer of the course may give an extension. It is for one student and one assignment, to a later date, with a reason. It is allowed until that student's submission is marked; the penalty is then recalculated.
BR-10	Only the HoD of the department that owns the course may moderate its marks. The person who entered a mark cannot moderate it.
BR-11	Only Moderated submissions can be released. After release only the HoD can correct the mark, with a reason.
BR-12	Every creation or change of a mark is recorded in the audit log with who, when, old value, new value and reason. The first mark has an empty old value and the reason "Initial marking".
BR-13	Accounts, courses and assignments are never deleted, only deactivated or closed, so that history and audit records stay valid.
BR-14	A mark is a number from 0 to the assignment's maximum mark in steps of 0.5.
BR-15	The administrator manages accounts and structure only and cannot view marks, feedback or submitted files.

Table 4.4: Permission matrix

Function	Admin	Lecturer	HoD	Student
Log in, change own password	✓	✓	✓	✓
Manage accounts, reset passwords, back up data	✓	—	—	—
Manage departments, courses, lecturer assignment, enrolment	✓	—	—	—
Create/edit/close assignments (own courses)	—	✓	—	—
Grant individual extension (own courses)	—	✓	—	—
View submissions and open files (own courses)	—	✓	✓ (own department)	own only
Mark and re-mark before moderation (own courses)	—	✓	—	—

Function	Admin	Lecturer	HoD	Student
Review, moderate, release, correct marks (own department)	—	—	✓	—
View audit log (own department)	—	—	✓	—
View courses and assignments	—	—	—	✓ (enrolled)
Submit and resubmit work	—	—	—	✓
View marks and feedback	—	✓ (own courses)	✓ (own department)	released only

5. Use Case Model

5.1 Actors

Actor	Type	Description
User (abstract)	Generalised actor	Any person logged in. It cannot exist on its own; it holds what all four roles share: log in and change password. The four roles below specialise it (generalisation).
Administrator	Primary, specialises User	Manages accounts, departments, courses, lecturer assignment, enrolment and backups. Cannot see marks (BR-15).
Lecturer	Primary, specialises User	Creates and manages assignments, grants individual extensions, marks submissions and writes feedback.
Head of Department	Primary, specialises User	Reviews and moderates marks for the department's courses, releases approved marks, corrects released marks and reads the audit log.
Student	Primary, specialises User	Views courses and assignments, submits and resubmits work, and reads released marks and feedback.

5.2 Use Case Diagram

Figure 5.1 shows all four actors. **Generalisation** is used for the User actor and for the specialisation of "Manage user accounts" into create and deactivate. **Include** is used for behaviour that always happens: validating a submitted file, writing feedback and recording

an audit entry when marking or moderating, and reviewing marks during moderation. **Extend** is used for optional or conditional behaviour, resubmission (only before the deadline), late submission (only after the effective deadline), adjusting or returning a mark during moderation, and locking an account after repeated failed logins.

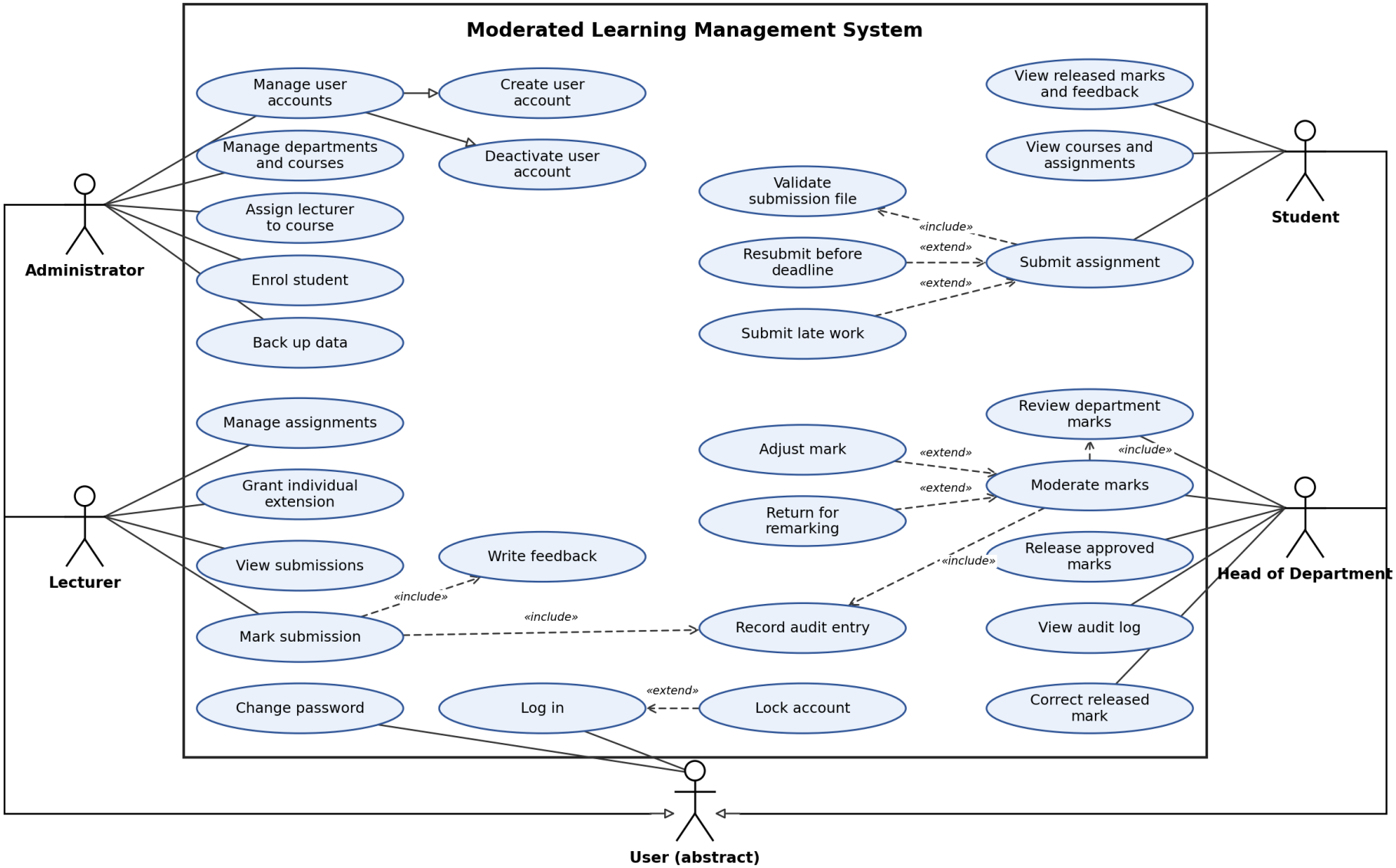


Figure 5.1: Use case diagram of the Moderated LMS (generalisation, include and extend

6. Risk Register Update at Gate 1

Table 6.1: Risk register at Gate 1 (L, I and exposure are the current values)

ID	Risk	L	I	Exp.	Level	Change at Gate 1	Mitigation	Owner
R1	Scope keeps changing	3	3	9	Medium	Lowered: all 7 conflicts resolved; Won't list fixed.	Change control; remove Should items first.	Student
R2	Work takes longer than expected	4	4	16	High	Unchanged. SRS has 10 Should items that can be cut.	Weekly Friday check; cut list in Section 4.4.	Student
R3	Illness, power cuts, computer failure	2	5	10	Medium	Unchanged.	Back up work; push to Git regularly.	Student
R4	Developer lacks some technical skills	3	3	9	Medium	Unchanged.	Prototype libsodium (Argon2id) and SQLite in Week 2–3.	Student
R5	Security problems	2	5	10	Medium	Unchanged; now measurable (NFR-05 to NFR-07).	Parameterised queries, salted hashes, role tests.	Student
R6	System is too slow	2	3	6	Low	Unchanged; limits set in NFR-01, NFR-02.	Indexes; test with 800 students.	Student
R7	Design, code and tests do not match	3	4	12	Medium	Unchanged; traceability started (Appendix A).	Keep the matrix updated at each gate.	Student

ID	Risk	L	I	Exp.	Level	Change at Gate 1	Mitigation	Owner
R8	Tool or library problems	3	3	9	Medium	Unchanged; libsodium must build on Windows and Ubuntu (NFR-13).	Set up tools and CI early; keep a fallback library.	Student
R9	Rules are unclear	2	3	6	Low	Lowered: rules written as BR-01 to BR-15.	Keep rules in the SRS only; change by control.	Student
R10	Stakeholders are unavailable	3	3	9	Medium	Unchanged.	Supervisor confirms Section 3 at Gate 1.	Student, Supervisor
R11	NEW: Stakeholders reject the conflict resolutions	3	4	12	Medium	Added at Gate 1.	Present Section 3 at Gate 1; record outcome; use change control.	Student, Supervisor
R12	NEW: Argon2id makes login too slow on old computers	3	3	9	Medium	Added at Gate 1 (from CF7).	Tune parameters; time NFR-01 on the oldest available machine in Week 4.	Student
R13	NEW: Errors in deadline and penalty logic (rounding, time zone, boundaries)	3	4	12	Medium	Added at Gate 1 (from AM1–AM3).	Isolate in one penalty class; boundary tests from Table 4.4; store UTC.	Student
R14	NEW: File handling failures (bad paths, large files, full disk)	3	2	6	Low	Added at Gate 1 (from FR-16).	Validate files; handle exceptions; failure tests (NFR-08).	Student